

Fernando Hernandez Cantu

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EDUCATION

Monterrey Institute of Technology and Higher Education, Campus Monterrey – ITESM

Aug 2024 – Jun 2028

Bachelor of Computer Engineering

GPA 95.34/100

CBTIS No. 36 Campus Monclova, Coah.

Aug 2022 – Jun 2024

Technical High School in Mechatronics – Awarded academic excellence

GPA 99/100

WORK EXPERIENCE

distritotec

Software Engineering Intern

Aug 2025 – Present

- Developing a project for estimating crowd size and improving public space efficiency by studying how ITESM's Central Park and Oasis Center is used;
- Developing backend services using **PostgreSQL**, **MongoDB**, **Docker Compose** within microservices architecture deployed on **Azure**;
- Implementing **video processing pipelines** using **Video-LLaMA**, **DeepStream**, and streaming via **FFmpeg** codecs.

PROJECTS

Robocup @Home

May 2025 - Present

Member of the computer vision and robotic manipulation teams in RoBorregos

- Optimized point clouds for 3D perception, reducing total points by 90% while enhancing resolution in key areas using **ROS 2** and **gazebo**;
- Developing object detection modules using **YOLO**, **OpenCV**, **ROS 2** and **Docker**;
- Optimizing Pipeline to train a **YOLO** model with image segmentation with **Grounding DINO** and **SAM** models;
- Implemented **ClearGrasp** to improve 3D perception of transparent and reflective objects.

Dashboard-Tec

Ago 2025 - Present

Full stack developer for a community web platform to discover and share events at ITESM.

- Developing the event submission, calendar and admin pages using **Prisma**, managing both **MySQL** and **PostgreSQL** databases;
- Developing backend services with **tRPC** and user **authentication**;
- Built the frontend using **TypeScript**, **React**, **Tailwind**, and **Next.js**.

SWE Training by RoBorregos

May 2025 – Aug 2025

*Frontend Developer for a web platform to manage **LeetCode** weekly problems, resources, and leaderboard rankings.*

- Development using **TypeScript**, **React**, **Tailwind**, **Next.js**;
- Provided a platform with 650+ solved problems, helping 35+ students strengthen technical skills for their professional growth.

Robocup Rescue Maze Jr.

Nov 2024 – May 2025

Developed an autonomous robot for navigating a maze and deploying rescue kits.

- Achieved **1st place** at **regional** competition and **2nd place** at the **Robotics Mexican Tournament** representing RoBorregos;
- Developed a computer vision system to detect victims using a **TensorFlow** model and **OpenCV** in **Python**;
- Developed an autonomous control and navigation system, implementing **PID controllers** in **C++** with **PlatformIO**;
- Implemented a serial communication protocol between a **Jetson Nano** and an **ESP32** to transmit **TensorFlow** model detection results;
- Led the project as team **captain**, coordinating development and competition.

Candidates by RoBorregos

Aug 2024 – Nov 2024

*Developed an autonomous robot, earned **1st place** in the annual RoBorregos contest.*

- Robot focused on navigating mazes, grabbing objects, and following lines;
- Implemented Arduino with multiple sensors such as encoders, **MPU gyroscope**, **infrared sensors**, **color sensor**, and **ultrasonic sensor**.

SKILLS

PROGRAMMING LANGUAGES

3 years: C++

2 years: Python

1 year: R, MATLAB, TypeScript, JavaScript, SQL

OTHER AWARDS AND COMPETITIONS

11th generation “Lideres del Mañana” ITESM (May 2024)

Bronze Medal in the National Mathematics Olympiad for Elementary and Middle School Students **ONMAPS** (2021)

Bronze Medal in the National Mathematics Olympiad for High School Students **ONMB** (2024)

Second Place in the National Academic Festival **DGTI** (May 2023)

TECHNOLOGIES

HTML, CSS, Tailwind, Node.js, React, Next.js, T3, Prisma, tRPC
PostgreSQL, SQLite, mySQL, Vercel, TensorFlow, PyTorch, OpenCV,
Git, Ubuntu, Bash, VS Code, Arduino, PlatformIO, ROS2, Gazebo,
Docker, YOLO, Proteus, EasyEDA, SolidWorks.

LANGUAGES

Spanish	Native
English	Advanced (C1)
French	Basic (A1-A2)